Construction manual · 1000x

Economical 1000x version · 12 V adapter, 100x oil objective

Part	Support with light for the pregoscope with a 100x/1.25 oil objective and a WF10X eyepiece (1000x)
Units	millimetres (mm)
Tolerance	±0.2 mm (ISO 2768-m; PLA, 0.2 mm layer)
Origin	on the tube axis, z = 0 at the top face of the lid, +z up, +x to the right seen from the front, +y toward the column
Support height	`237.6 mm`
Estimated time	165 min of assembly, not counting printing

Status: pre-calibration, no prototype

- Pre-calibration and no prototype: nothing of this version has been built yet. The dimensions marked "suposto" (assumed) in the table of section 3 must be measured on the purchased parts before printing in series.
- Pregoscope tube: from the objective shoulder to the rim it is 152 mm; the 160/0.17 objective was calculated for 160 mm, and 155 mm is the most this tube can give.
- The 100x objective was assumed to have a 45 mm parfocal distance (Bresser, Optika); the 40x bought measures 35 mm. Measure the 100x you buy: its length sets the height of the support (237.6 mm).
- The temperature of the heatsink inside the PLA box is an estimate (0.7 to 1.0 W of heat at 350 mA) and was not measured. Check it with test point 6.
- The driver is one module for 12 V in and 350 mA out. I did not find one on sale in Brazil: the alternative is a 2.8 to 6 V constant-current module behind a step-down.
- This instrument is not diagnostic equipment and has not been validated against a reference microscope.
- The Chinese and English text was written without a technical translator and needs review by a native speaker.

Revisions

Rev.	Date	Description
B	2026-11-08	First issue of the economical 1000x version, pre-calibration

Before you start

Tools and materials

- FDM 3D printer, bed of at least 300 mm, PLA filament (black PLA for the light tube)
- 30 to 40 W soldering iron, 0.8 mm solder, wire stripper, heat-shrink
- Multimeter with a 10 A input (the mA input has a small fuse); contact or infrared thermometer
- Thermal paste, hot-glue gun, double-sided tape
- Small Phillips screwdriver; caliper and micrometer; lens paper and isopropyl alcohol

Safety

- Soldering iron at about 300 °C, hot glue and heat-shrink burn: use the stand, ventilate and let them cool.
- Do not look straight at the lit 3 W LED and do not power it without the heatsink: it burns out in seconds.
- Measure the LED current on the 10 A input of the multimeter, in series; the mA input has a small fuse and blows at 350 mA.
- Check the polarity before switching on (D2, LED, plug with positive centre). The driver has no reverse-polarity protection.
- Use only the 12 V adapter indicated. The heatsink and the driver get hot: do not touch right after switching off.
- Immersion oil: follow the maker's sheet, avoid contact with skin and eyes and wipe any drop that falls into the box.
- Do not overtighten screws in PLA and do not force the tip of the objective against the cover glass.

Printing

PLA · 0.2 mm · 3 perimeters · 20 to 30 % infill (wheel AP1 at 100 %). Dark walls help to reduce stray light.

ID	Part	Orientation
M1	Support: light box, column and arm	file -impressao, lying on its back; supports only inside the box and at the wheel slot
M2	Tube clamp	file -impressao, opening upward; no supports (the clamp of this version sits higher than the 400x one)
M3	Bottom lid	already flat; no supports; the 12 air holes lie around the heatsink
M4	Diffuser	flat, natural or white PLA, 1 to 2 layers; no supports
M5	Light tube with wheel pin (printed, black)	standing, black PLA, the foot slot to the side; no supports; the pin at the top is thin: print slowly
AP1	Aperture wheel (printed)	file -impressao, flat; 3 perimeters and 100 % infill; no supports

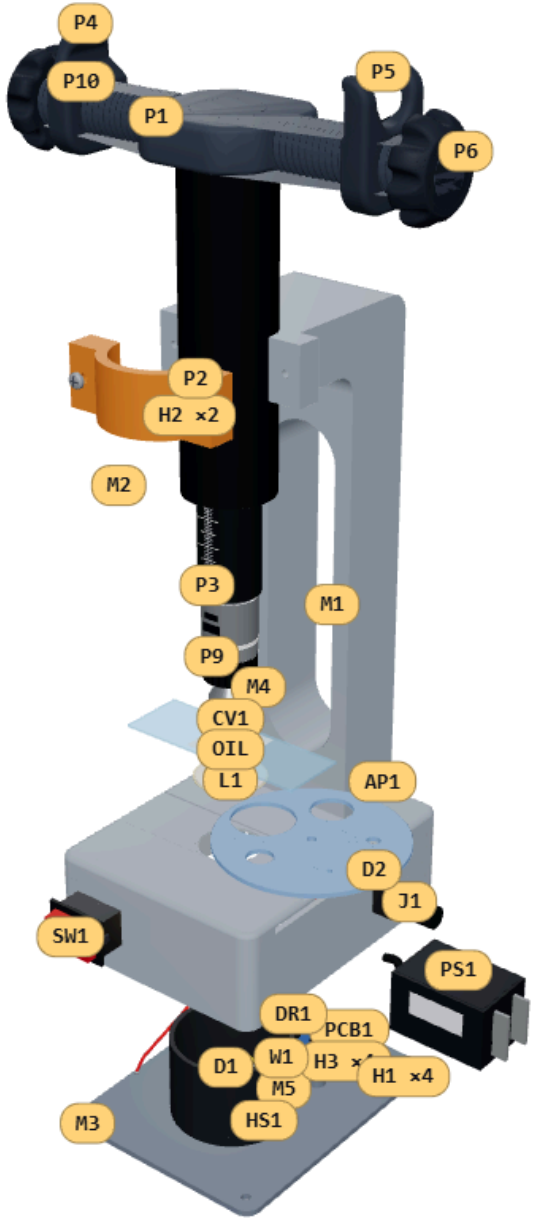
Files that come with this manual

File	Content
`mecanica/v2-1000x/suporte-base-coluna-impressao.stl`	M1 support, in the print orientation
`mecanica/v2-1000x/suporte-base-coluna.stl`	M1 support, in the assembly position

File	Content
mecanica/v2-1000x/abracadeira-tubo-impressao.stl	M2 clamp, in the print orientation (higher than the 400x one)
mecanica/v2-1000x/tampa-fundo.stl	M3 bottom lid, with 12 air holes
mecanica/comum/difusor.stl	M4 diffuser
mecanica/v2-1000x/tubo-de-luz.stl	M5 light tube with wheel pin and wire slot
mecanica/v2-1000x/roda-de-aberturas-impressao.stl	AP1 aperture wheel, flat
eletronica/v2-1000x-esquematico.svg	Schematic of the 1000x version
CUSTOS-v2-1000x-1un.csv	Cost of the 1000x version, 1 unit (R$, Brazil)
mecanica/calibracao/	Jigs G1 to G3 (print them first)

Order of the document: exploded view · 1 drawings and dimensions · 2 positions · 3 bill of materials · 4 wiring · 5 assembly · 6 test · 7 optics and calibration · 8 inspection.

Exploded view



Every part pulled apart with its ID; "×n" is the quantity.

1. Drawings and dimensions

For this version the STL files are the reference of the printed parts. The technical drawings DT-01 to DT-09 of the v2 (400x) manual hold for the clamp and the general shape, but do not show the 237.6 mm height, the wheel slot in the right wall, the jack hole at y = 32 mm or the light tube: use the STL files and the table below, which comes from the same numbers that generate the STL files.

ID	Part	Dimensions (mm)	Source
P9	100x/1.25 oil objective (160/0.17, RMS, spring tip)	⌀ 20,0 × 44,8 mm sob o ombro (+ 3 de rosca RMS), ponta com mola	assumed: measure
CV1	Cover glass 22 × 22 × 0.17 mm	22 × 22 × 0,17 mm	from a page
OIL	Microscope immersion oil (100 ml)	⌀ 5 × 0,2 mm	from a page
L1	Microscope slide	76 × 26 × 1 mm	from a page
M4	Diffuser (printed)	⌀ 26,0 × 0,8 mm	measured
AP1	Aperture wheel (printed) (printed)	⌀ 72,0 × 1,8 mm, furos de 24,0, 10,0, 2,0, 5,0, 16,0 mm num círculo de raio 22,0 mm	calculated
M5	Light tube with wheel pin (printed, black) (printed)	⌀ 48,4 ext. / 44,4 int. × 34,2 mm, base de ⌀ 10,0 mm e pino de ⌀ 3,0 mm em x = 22,0 mm	calculated
D1	3 W cool-white LED on an aluminium star board	estrela ⌀ 20,0 × 1,6 mm, emissor ≈ ⌀ 6 × 3,2 mm	assumed: measure
HS1	Round aluminium heatsink	⌀ 30,0 × 12,0 mm	assumed: measure
DR1	Constant-current driver, 350 mA, 12 V input	30 × 20 × 10 mm	assumed: measure
PCB1	Perforated board 40 × 25 mm (carries the driver)	40 × 25 × 1,6 mm, furos M3 a 3,5 mm das bordas	measured
D2	Diode 1N5819 (reverse polarity)	DO-41, 5 × ⌀ 2,7 mm	from a page
SW1	Rocker switch KCD1-101	recorte 21,2 × 15,0 mm	from a page
J1	P4 panel jack DC-022B	furo ⌀ 7,8 mm	from a page
PS1	12 V 1 A switching adapter, P4 plug, positive centre	plugue P4 5,5 × 2,1 mm, centro positivo	from a page

Box and column of this version

Box	`90 × 115 × 40`
Total height	`237.6`
Hole of the jack J1	`⌀ 7.8, y = 32, z = 20`

Wheel slot (right wall)	`61.0 × 2.6, z = 36.1`
LED window and diffuser	`⌀ 24 · ⌀ 26`
Slide channel	`26.6 × 0.6`

2. Position of each part

Centre of the part in the coordinate system of the cover (mm). “Inside” means the part is inside the closed box.

ID	Name	X	Y	Z	Size X × Y × Z	Fixing	Inside
M1	Support: light box, column and arm	0	12.5	118.8	90 × 115 × 237.6	standing on the table	no
M3	Bottom lid	0	12.5	-1.2	90 × 115 × 2.4	4 screws H1 at the corners	yes
M5	Light tube with wheel pin (printed, black)	0	0	17.1	60.4 × 48.4 × 34.2	standing on the lid, hot glue at the base, foot slot toward the column (+y)	yes
HS1	Round aluminium heatsink	0	0	6	30 × 30 × 12	at the centre of the lid, inside the light tube, hot-glued at the base	yes
D1	3 W cool-white LED on an aluminium star board	0	0	12.8	20 × 20 × 4.8	star board on the heatsink, with thermal paste (TP1)	yes
PCB1	Perforated board 40 × 25 mm (carries the driver)	0	42	6.8	40 × 25 × 1.6	4 screws H3 on the 6 mm standoffs	yes
DR1	Constant-current driver, 350 mA, 12 V input	0	42	12.6	30 × 20 × 10	glued on PCB1 (hot glue or double-sided tape); inputs on the left, outputs on the right	yes
D2	Diode 1N5819 (reverse polarity)	23.6	41.7	17	2.7 × 13 × 2.7	anode soldered to the + lug of J1, body in the air along y, band toward the free end	yes
SW1	Rocker switch KCD1-101	-25	-43.8	20	21.2 × 10 × 15	front-wall cut-out, snap tabs	no
J1	P4 panel jack DC-022B	43.8	32	20	12.4 × 7.8 × 7.8	right-wall hole (behind the light tube), nut outside	no
PS1	12 V 1 A switching adapter, P4 plug, positive centre	85	32	20	38 × 26 × 24	outside the box, cable to the jack J1	no
M4	Diffuser	0	0	38.2	26 × 26 × 0.8	hot glue (W3) in the ceiling seat	yes
AP1	Aperture wheel (printed)	22	0	36.1	72 × 72 × 1.8	slides in from the side through the right-wall slot and turns on the pin of M5	yes
L1	Microscope slide	0	0	39.7	76 × 26 × 1	slid into the channel from the side	no
CV1	Cover glass 22 × 22 × 0.17 mm	0	0	40.3	22 × 22 × 0.2	on the slide, over the sample	no
OIL	Microscope immersion oil (100 ml)	0	0	40.5	5 × 5 × 0.2	one drop between the cover glass and the objective	no
H1	M3 × 10 self-tapping screw	0	0	-1.2	5.6 × 5.6 × 2	4 lid holes (see the table below)	yes
H2	M3 × 16 self-tapping screw	0	-12.8	226.6	54.4 × 16 × 5.6	2 clamp ears, Y axis (see the table below)	no

ID	Name	X	Y	Z	Size X × Y × Z	Fixing	Inside
H3	M3 × 8 self-tapping screw	0	42	7.6	5.6 × 5.6 × 2	4 PCB1 holes (see the table below)	yes
M2	Tube clamp	0	-10.7	226.6	59.7 × 20.3 × 22	2 screws H2 on the ears	no
W1	24 AWG flexible wire, 5 wires (see the cut list)	0	10	14	80 × 80 × 20	5 wires F1 to F5 between the parts, along the paths of the cut list	yes
W2	Heat-shrink tube ⌀2-3 mm	0	24	5	10 × 10 × 4	over the joints, 1 cm each	yes
W3	Hot glue and double-sided tape	0	0	38	26 × 26 × 1	bead around the diffuser edge	yes
P9	100x/1.25 oil objective (160/0.17, RMS, spring tip)	0	0	63.2	20 × 20 × 44.8	screwed to the inner tube of the pregoscope	no
P2	Outer tube	0	0	180.1	33.5 × 33.5 × 115	in the column ring, held by M2	no
P1	Head: plate and threaded rod	0	0	245.1	50 × 50 × 15	resting on top of the ring	no

Screws, one by one

ID	Screw	X	Y	Z
H1	M3 × 10 self-tapping screw	-37.6	-37.6	-2.4
H1	M3 × 10 self-tapping screw	-37.6	62.6	-2.4
H1	M3 × 10 self-tapping screw	37.6	-37.6	-2.4
H1	M3 × 10 self-tapping screw	37.6	62.6	-2.4
H3	M3 × 8 self-tapping screw	-16.5	33	6
H3	M3 × 8 self-tapping screw	-16.5	51	6
H3	M3 × 8 self-tapping screw	16.5	33	6
H3	M3 × 8 self-tapping screw	16.5	51	6
H2	M3 × 16 self-tapping screw	-23.8	-12.8	226.6
H2	M3 × 16 self-tapping screw	23.9	-12.8	226.6

3. Bill of materials (BOM)

ID	Name	Qty	Size	Type
M1	Support: light box, column and arm	1	237,6 mm	printed
M2	Tube clamp	1	Anel de 22 mm de altura, furo de 33,7 mm, dois parafusos M3 × 16	printed
M3	Bottom lid	1	Placa de 2,4 mm com porta-LED; v1 leva o berço da pilha, v2 leva 4 suportes da placa do dimmer	printed
M4	Diffuser	1	Disco de ⌀ 26 × 0,8 mm em PLA natural ou branco	printed
M5	Light tube with wheel pin (printed, black)	1	⌀ 48,4 ext. / 44,4 int. × 34,2 mm, base de ⌀ 10,0 mm e pino de ⌀ 3,0 mm em x = 22,0 mm	printed
AP1	Aperture wheel (printed)	1	⌀ 72,0 × 1,8 mm, furos de 24,0, 10,0, 2,0, 5,0, 16,0 mm num círculo de raio 22,0 mm	printed
P1	Head: plate and threaded rod	1	50 × 15 mm, haste de 170 mm	supplied (pregoscope)
P2	Outer tube	1	⌀ 33,5 mm e 115 mm de comprimento, medidos com paquímetro (o STEP V1 dizia 30 mm)	supplied (pregoscope)
P3	Inner tube (telescopic)	1	⌀ 22,5 mm; sai até 40 mm do tubo externo; escala em mm impressa no tubo	supplied (pregoscope)
P4	Left jaw	1	10 mm de espessura	supplied (pregoscope)
P5	Right jaw	1	10 mm de espessura	supplied (pregoscope)
P6	Right knob	1	⌀ 34 mm	supplied (pregoscope)
P10	Left knob (mirror of P6)	1	⌀ 34 mm, igual à P6	supplied (pregoscope)
P9	100x/1.25 oil objective (160/0.17, RMS, spring tip)	1	⌀ 20,0 × 44,8 mm sob o ombro (+ 3 de rosca RMS), ponta com mola	purchased
D1	3 W cool-white LED on an aluminium star board	1	estrela ⌀ 20,0 × 1,6 mm, emissor ≈ ⌀ 6 × 3,2 mm	purchased
HS1	Round aluminium heatsink	1	⌀ 30,0 × 12,0 mm	purchased
DR1	Constant-current driver, 350 mA, 12 V input	1	30 × 20 × 10 mm	purchased
D2	Diode 1N5819 (reverse polarity)	1	Schottky 1 A 40 V, DO-41, em série no positivo do conector J1	purchased

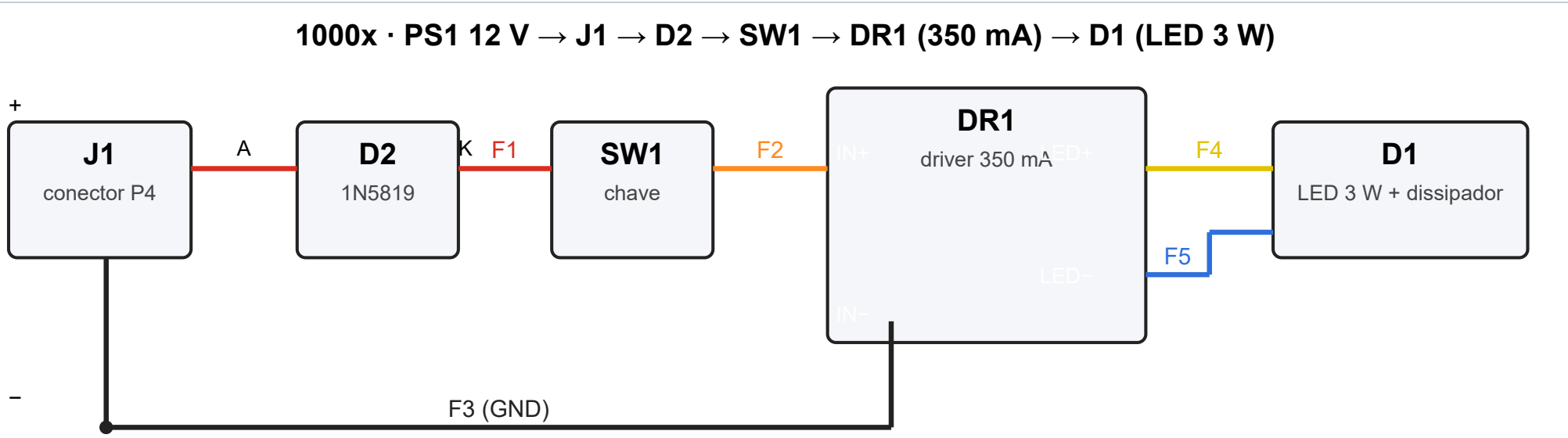
ID	Name	Qty	Size	Type
SW1	Rocker switch KCD1-101	1	2 terminais, liga-desliga, recorte de painel de 21,2 × 15,0 mm	purchased
J1	P4 panel jack DC-022B	1	furo ⌀ 7,8 mm	purchased
PS1	12 V 1 A switching adapter, P4 plug, positive centre	1	plugue P4 5,5 × 2,1 mm, centro positivo	purchased
PCB1	Perforated board 40 × 25 mm (carries the driver)	1	40 × 25 × 1,6 mm, furos M3 a 3,5 mm das bordas	purchased
H1	M3 × 10 self-tapping screw	4	M3 × 10	purchased
H2	M3 × 16 self-tapping screw	2	M3 × 16	purchased
H3	M3 × 8 self-tapping screw	4	M3 × 8	purchased
W1	24 AWG flexible wire, 5 wires (see the cut list)	1		purchased
W2	Heat-shrink tube ⌀2-3 mm	1	10 cm, uma luva de 1 cm por emenda	purchased
W3	Hot glue and double-sided tape	1	pontos de cola de 3 a 4 mm; fita de 12 mm	purchased
TP1	Thermal paste	1	—	purchased
L1	Microscope slide	1	76 × 26 × 1 mm, vidro	purchased
CV1	Cover glass 22 × 22 × 0.17 mm	1	22 × 22 × 0,17 mm	purchased
OIL	Microscope immersion oil (100 ml)	1	nD ≈ 1,518	purchased

Parts P1 to P6 and P10 are the pregoscope and the WF10X eyepiece is supplied by the customer; the 100x objective (P9) is bought. Check every part marked "suposto" (assumed) with the caliper before printing in series.

Measurements to take on the purchased parts

ID	Part	Value used in the 3D (mm)	Measured
P9	100x/1.25 oil objective (160/0.17, RMS, spring tip)	⌀ 20,0 × 44,8 mm sob o ombro (+ 3 de rosca RMS), ponta com mola	
D1	3 W cool-white LED on an aluminium star board	estrela ⌀ 20,0 × 1,6 mm, emissor ≈ ⌀ 6 × 3,2 mm	
HS1	Round aluminium heatsink	⌀ 30,0 × 12,0 mm	
DR1	Constant-current driver, 350 mA, 12 V input	30 × 20 × 10 mm	

4. Schematic and wiring



D2: ânodo soldado direto no terminal + do J1 (sem fio). O ânodo do D2 fica no lado sem faixa.
Cores dos fios: F1 vermelho · F2 laranja · F3 preto · F4 amarelo · F5 azul

Soldering order

1. Solder F4 (yellow) to the + pad and F5 (blue) to the − pad of the LED star, going through the foot slot of the light tube.
2. Solder the anode of D2 (the end without the band) straight to the + lug of the jack J1.
3. Solder F1 (red) to the cathode of D2 and to terminal 1 of the switch SW1.
4. Solder F2 (orange) to terminal 2 of the switch and to the input IN+ of the driver.
5. Solder F3 (black) to the − lug of J1 and to the input IN− of the driver.
6. Solder the other ends of F4 and F5 to the outputs LED+ and LED− of the driver.
7. Protect every joint with a 1 cm piece of heat-shrink and check that no wire touches the heatsink.

Circuit connections

1. Positivo da fonte: + lug of the jack J1 = anode of D2 (end without the band)
2. Depois do D2: cathode of D2 (banded end) = terminal 1 of the switch SW1
3. Positivo depois da chave: terminal 2 of the switch SW1 = input IN+ of the driver DR1
4. Negativo da fonte: − lug of the jack J1 = input IN− of the driver DR1
5. Saída positiva do driver: output LED+ of the driver DR1 = pad + of the LED D1 (red)

6. Saída negativa do driver: output LED− of the driver DR1 = pad − of the LED D1 (blue)

Wire cut list

ID	Colour	From	To	Path (mm)	Cut to (mm)
F1	red	D2 K	SW1 1	102	**185**
F2	orange	SW1 2	DR1 IN+	81	**165**
F3	black	J1 -	DR1 IN-	45	**130**
F4	yellow	DR1 LED+	D1 +	75	**105**
F5	blue	DR1 LED-	D1 -	57	**85**

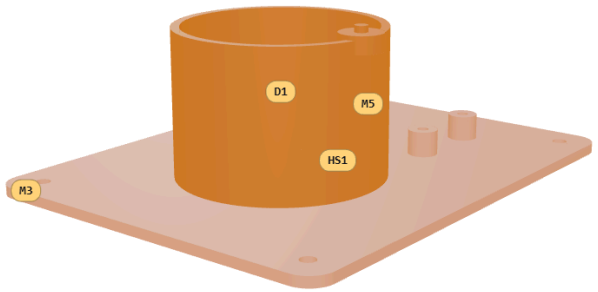
Cut = path + 12 mm of stripped ends (6 mm each side) + slack (70 mm on wires F1 to F3, which start on parts of the wall, so the lid opens without disconnecting anything; 15 mm on the LED wires F4 and F5). 24 AWG flexible wire. The paths were calculated around the parts and checked by a test.

There is no custom circuit board in this version: the driver comes ready-made and the perforated board is only a carrier. If you use two modules instead of one (a step-down and a 2.8 to 6 V constant-current module), put the step-down between the switch and the module, set to about 5 V.

1

Put the LED on the heatsink and the light tube on the lid (M3, M5, HS1, D1, TP1)

15 min



Spread a thin layer of thermal paste (TP1) on the underside of the LED star (D1) and set it on the centre of the heatsink (HS1), which is hot-glued to the centre of the lid (M3). The two pads of the star (+ red, − blue) face the front. Put the light tube (M5) around it, with the foot slot toward the column (+y), and hot-glue its base.

⚠
Do not power the LED without the heatsink.

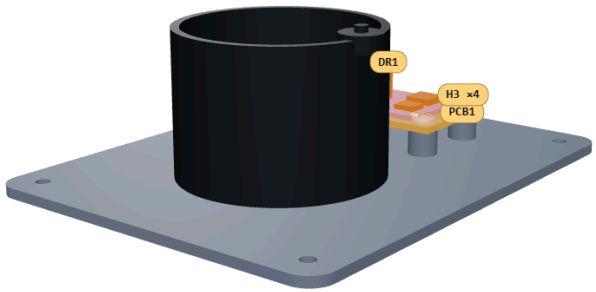
✓
heatsink at the centre; tube slot toward the column

ID	Part in this step	Qty
M3	Bottom lid	1
M5	Light tube with wheel pin (printed, black)	1
HS1	Round aluminium heatsink	1
D1	3 W cool-white LED on an aluminium star board	1

2

Fix the board PCB1 and the driver DR1 (PCB1, DR1, H3)

10 min



Screw the 40 × 25 mm perforated board (PCB1) on the 4 standoffs of 6 mm with 4 screws M3 × 8 (H3), behind the light tube. Glue the driver (DR1) at the centre of the board, with the inputs (IN+, IN−) to the left and the outputs (LED+, LED−) to the right.

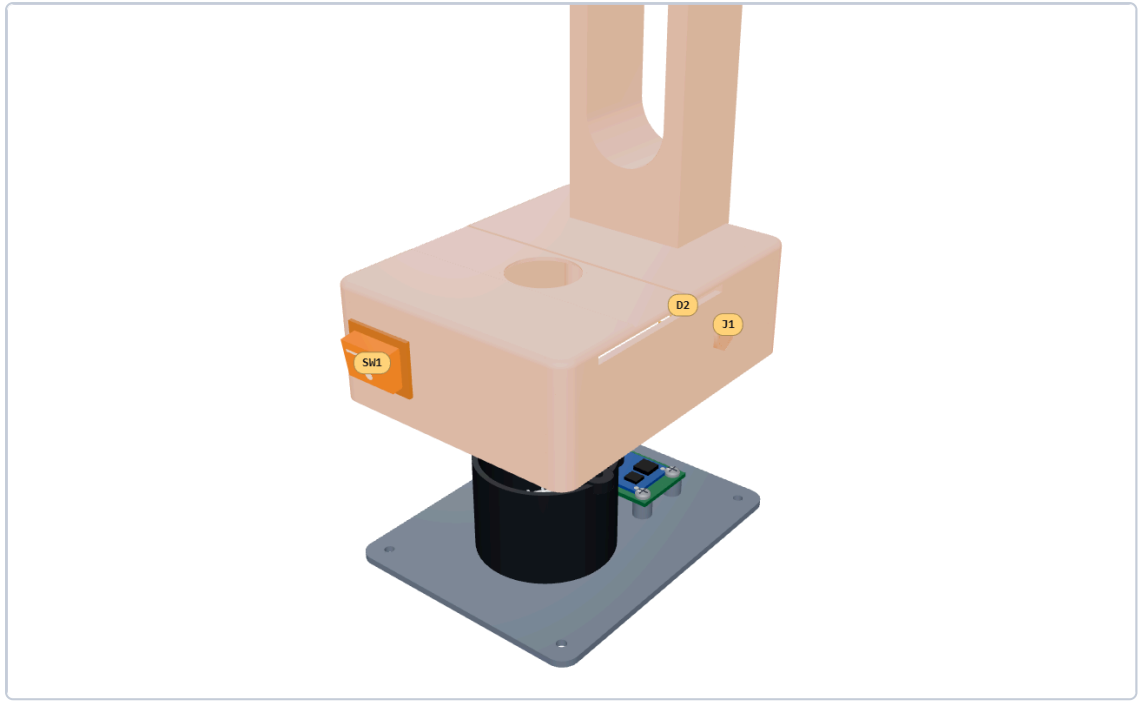
✓
the driver does not touch the light tube

ID	Part in this step	Qty
PCB1	Perforated board 40 × 25 mm (carries the driver)	1
DR1	Constant-current driver, 350 mA, 12 V input	1
H3	M3 × 8 self-tapping screw	4

3

Install the jack J1, the switch SW1 and the diode D2 in the box (M1, SW1, J1, D2)

15 min



With the box (M1) upside down, snap the switch SW1 into the front-wall cut-out and fit the jack J1 in the right-wall hole (behind the light tube, at y = 32 mm), nut outside. Solder the anode of the diode D2 (the end without the band) to the + lug of J1; D2 stands in the air along y, band toward the free end.

⚠
A reversed diode blocks the current: the band goes on the side toward the switch.

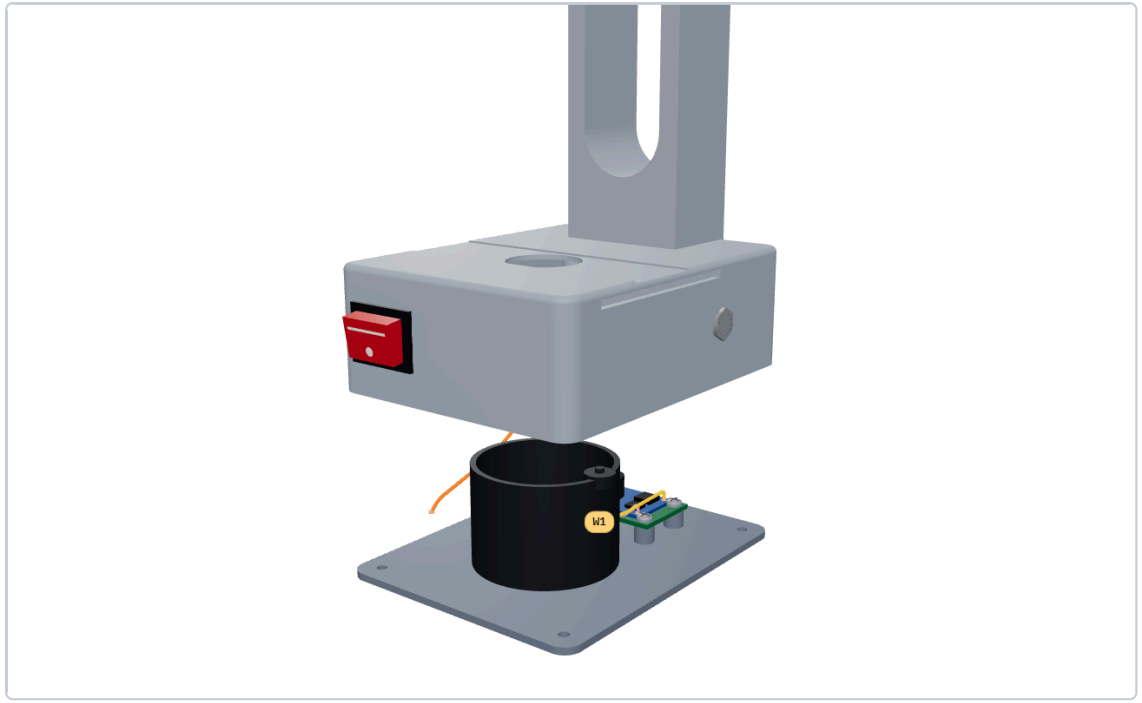
✓
the band of D2 is on the cathode (free end)

ID	Part in this step	Qty
M1	Support: light box, column and arm	1
SW1	Rocker switch KCD1-101	1
J1	P4 panel jack DC-022B	1
D2	Diode 1N5819 (reverse polarity)	1

4

Cut and solder the five wires F1 to F5 (W1, W2)

30 min



Cut the five wires to the lengths of the cut list and strip 6 mm from each end. Solder in the order of the soldering list: F4 and F5 (yellow and blue) go through the foot slot of the light tube to the pads of the star; F1 joins the cathode of D2 to the switch, F2 joins the switch to the input IN+ of the driver, F3 joins the − of J1 to the input IN−. Protect the joints with heat-shrink (W2).

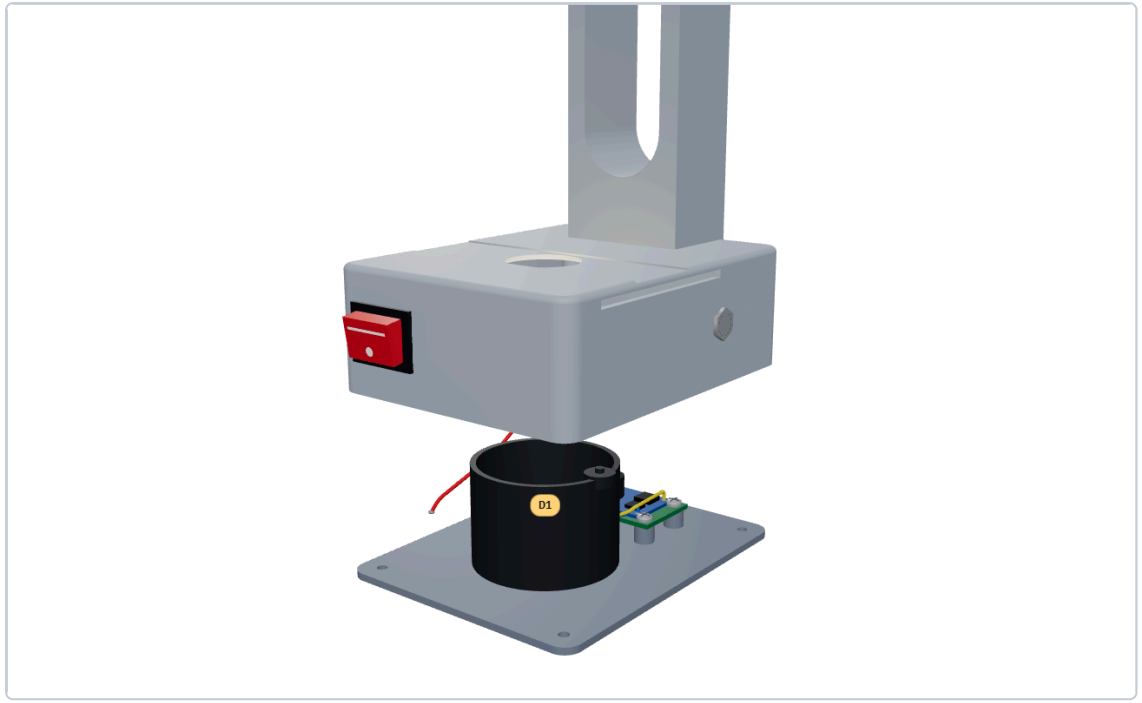
✓
each wire in the colour and on the pad of the list; no wire touches the heatsink

ID	Part in this step	Qty
W1	24 AWG flexible wire, 5 wires (see the cut list)	1
W2	Heat-shrink tube ⌀2-3 mm	1

5

Electrical test on the bench, with the lid open

40 min



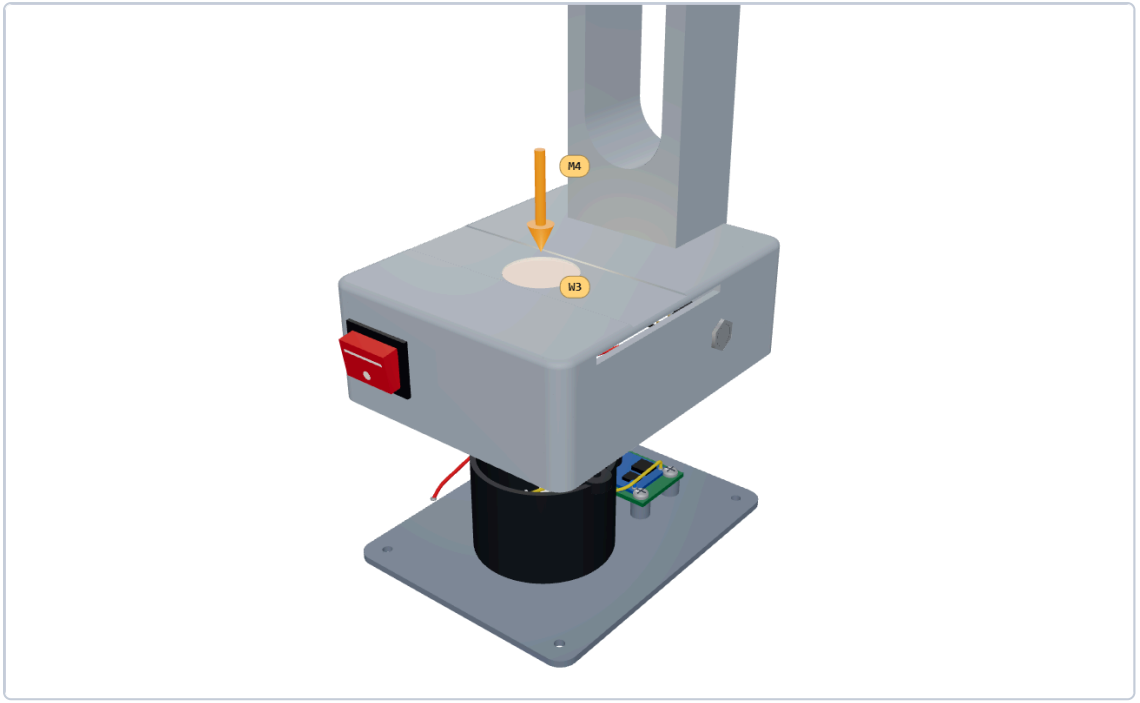
Before closing, switch the power on and follow the test plan of section 6 point by point: supply voltage, after D2, driver input, voltage across the LED, LED current on the 10 A input. Leave the LED on for 30 minutes and measure the temperature of the heatsink. If a reading is out of range, stop and fix it before going on.

✓
350 mA in the LED and heatsink up to 50 °C

6

Glue the diffuser (M4, W3)

10 min



From inside the box, set the diffuser (M4) in the seat of the ceiling, over the window of the LED, and run a bead of hot glue (W3) on its edge. The glue must not cover the window.

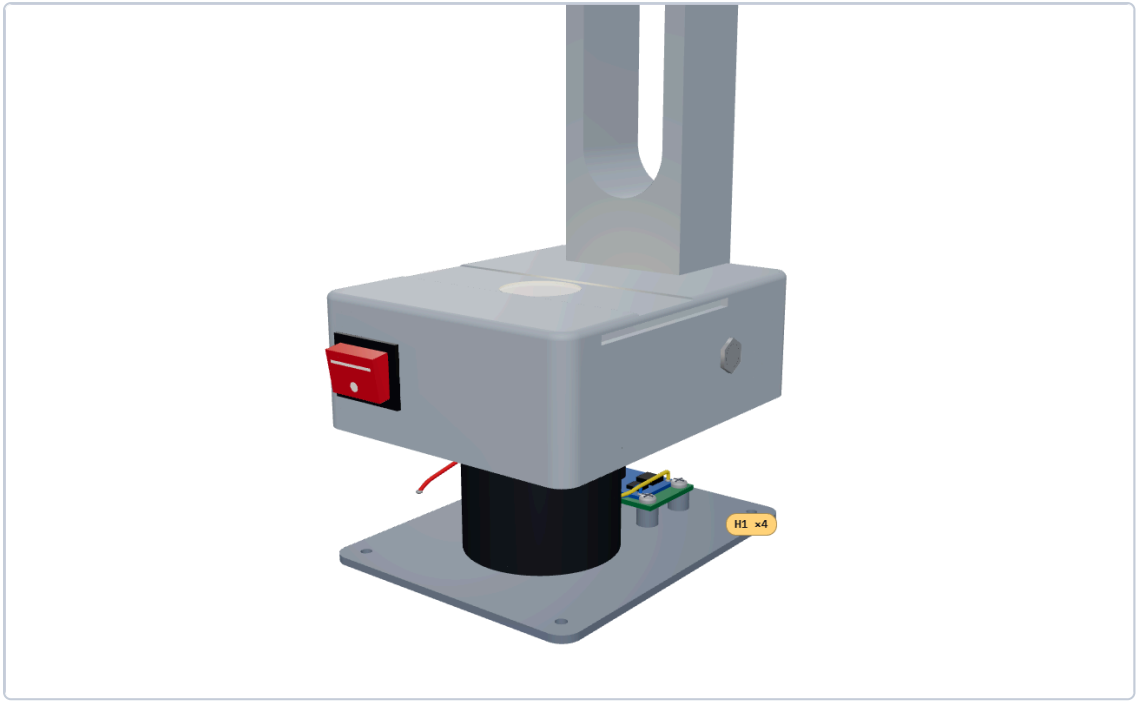
✓
the diffuser does not rock and the window is free

ID	Part in this step	Qty
M4	Diffuser	1
W3	Hot glue and double-sided tape	1

7

Close the lid (H1)

10 min



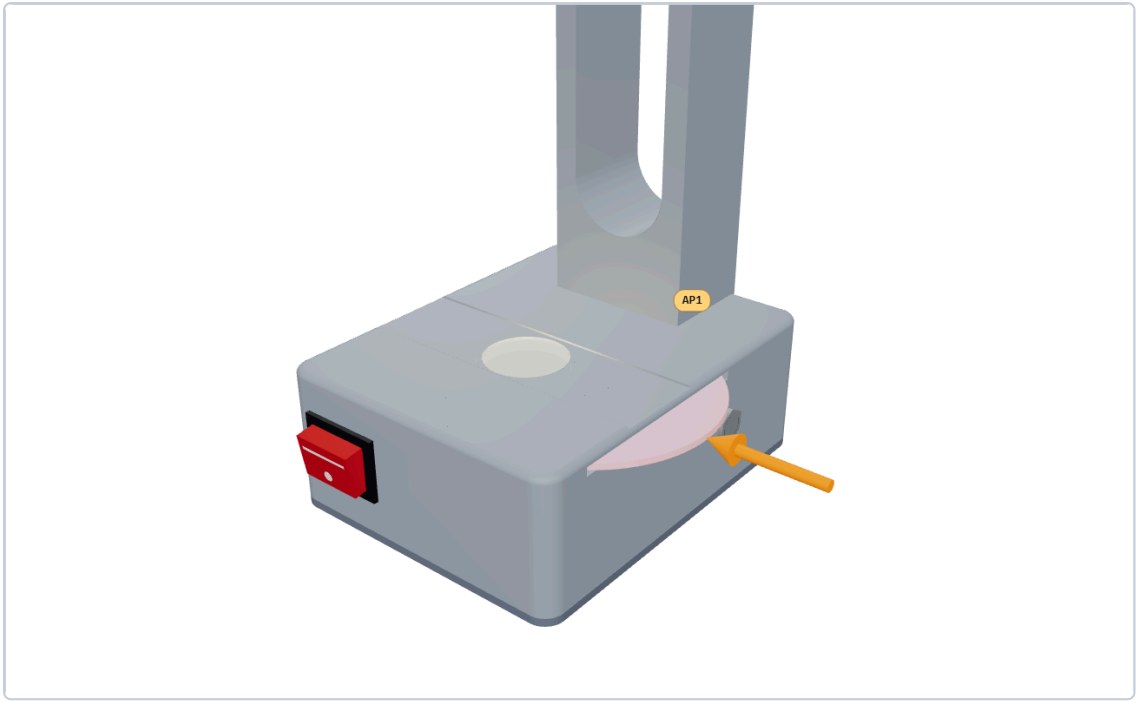
Raise the lid to the box without trapping any wire (leave the slack of wires F1 to F3) and tighten the 4 screws M3 × 10 (H1) at the corners. Tighten a little at a time, in a cross pattern, without forcing the plastic.

ID	Part in this step	Qty
H1	M3 × 10 self-tapping screw	4

8

Fit the aperture wheel (AP1)

5 min



With the lid closed, slide the wheel (AP1) in sideways through the slot of the right wall, edge first, until the central hole drops onto the pin of the light tube (M5). Turn the wheel: each of the five holes (24, 10, 2, 5 and 16 mm) must stop on the axis of the window, and the edge must stick out about 13 mm from the box.

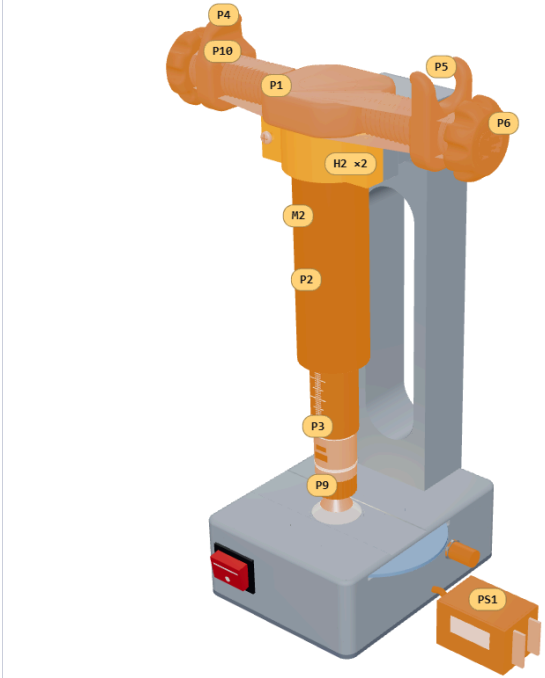
✓
the wheel turns without rubbing and no hole sits half on the window

ID	Part in this step	Qty
AP1	Aperture wheel (printed)	1

9

Fit the pregoscope with the 100x objective (M2, H2, P1 to P10, PS1)

15 min



Screw the 100x objective (P9) on the tip of the inner tube. Put the outer tube (P2) in the ring of the column and close the clamp (M2) with 2 screws M3 × 16 (H2) without overtightening. The WF10X eyepiece (supplied by the customer) goes on the rim of the outer tube. Plug the 12 V adapter (PS1) into the jack J1.

✓
152 mm from the objective shoulder to the rim

ID	Part in this step	Qty
M2	Tube clamp	1
H2	M3 × 16 self-tapping screw	2
P1	Head: plate and threaded rod	1
P2	Outer tube	1
P3	Inner tube (telescopic)	1
P4	Left jaw	1
P5	Right jaw	1

ID	Part in this step	Qty
P6	Right knob	1
P10	Left knob (mirror of P6)	1
P9	100x/1.25 oil objective (160/0.17, RMS, spring tip)	1
PS1	12 V 1 A switching adapter, P4 plug, positive centre	1

10

Put the slide, the cover glass and the oil, and focus (L1, CV1, OIL)

15 min



Cover the sample of the slide (L1) with the 0.17 mm cover glass (CV1) and slide the slide into the channel. Put a drop of oil (OIL) of about 5 mm on the cover glass. With the wheel at 24 mm, pull the inner tube out to 37 mm on the scale and go down slowly until the oil touches the objective; focus by turning the tube, without forcing the spring tip. Then reduce the opening of the wheel to adjust brightness and contrast.

⚠
Clean the objective and the cover glass with lens paper and isopropyl alcohol after each use.

✓
sharp image at 1000x and no oil in the box

ID	Part in this step	Qty
L1	Microscope slide	1

ID	Part in this step	Qty
CV1	Cover glass 22 × 22 × 0.17 mm	1
OIL	Microscope immersion oil (100 ml)	1

6. Electrical test

Supply: the 12 V adapter, 3 W cool-white LED at 350 mA. Measure in this order, with the lid open.

Measuring point	Expected	Range	If out of range	Measured
Supply at J1: positive to negative, no load	`12.0 V`	`11.5 to 15.0 V`	Weak adapter (under 11.5 V) or unregulated (up to 15 V unloaded): change the adapter.	
Cathode of D2 to GND, switch open	`leitura 1 – 0.1 a 0.6 V`	`11.0 to 12.3 V`	0 V: D2 reversed or a cold joint on J1; a drop over 0.6 V: wrong diode.	
Driver input IN+ to GND, switch on	`igual à leitura 2 (± 0.2 V)`	`10.8 to 12.3 V`	Switch without contact or a cold joint on wires F1 and F2.	
Voltage across the LED: LED+ to LED–	`3.2 V`	`3.0 to 3.6 V`	0 V or equal to the input: LED reversed or open; over 3.6 V: the driver is not limiting the current.	
LED current, multimeter in series on the 10 A input	`350 mA`	`330 to 370 mA`	Over 370 mA: a 700 mA driver or badly set, switch off; under 330 mA: driver set wrong or weak adapter.	
Heatsink temperature after 30 minutes lit, lid closed	`até 50 °C`	`até 50 °C`	Over 55 °C the PLA of the box softens: reduce the current, ventilate more or print in PETG.	

The ranges of the LED voltage and current and the temperature limit are design values, not standard ones. If a reading is out of range, stop and fix it before closing the box.

7. Optics, focus and calibration

Total magnification	1000x (100x × 10x)
Objective aperture	NA 1,25 (oil)
Theoretical resolution with the diffuser	0.31 µm
Field of view	0.18 mm
Tube, objective shoulder to rim	152 mm of 160 mm
Working distance	0.18 mm + 0,17 mm cover glass
Light	1.2 W electrical, 0.7–1.0 W of heat

1. Measure and note in the sheet below every part marked "suposto": objective (diameter and length from the shoulder to the tip), LED star, heatsink, driver, cover glass (micrometer).
2. Height: the 100x objective sets the height of the support (parameters objectiveLength and workingDistance). If the shoulder-to-tip length is not 44.8 mm, remake M1 and M2 with the measured value.
3. Focus: slide with a 0.17 mm cover glass, one drop of oil of about 5 mm, aperture wheel at 24 mm; lower the inner tube slowly until the oil touches the objective and focus by turning the tube.
4. Apertures: 24 mm gives the most light; 10 and 5 mm raise the contrast; 2 mm only to see edges. The wheel does not replace a condenser: if the contrast is not enough, an Abbe condenser can be added later without remaking the support.
5. Cleaning: after each use, lens paper and isopropyl alcohol on the objective and the cover glass.

After measuring the parts, update the parameters in holder.ts and suporte-1000x.ts, regenerate the files and issue revision B.

8. Final inspection

#	Check	OK	Not OK
1	The slide L1 with the cover glass CV1 moves in the channel without effort and without visible side play.	☐	☐
2	The pregoscope tube fits the ring and is held by M2 without cracking the PLA.	☐	☐
3	With the inner tube pulled out 37 mm on the scale, the 100x objective sits about 0.18 mm from the cover glass and the image focuses with one drop of oil.	☐	☐
4	The LED lights and the light is even over the window, with no hot spot.	☐	☐
5	LED current measured between 330 and 370 mA; voltage across the LED between 3.0 and 3.6 V.	☐	☐
6	Heatsink at 50 °C at most after 30 minutes with the LED on and the lid closed.	☐	☐
7	The aperture wheel turns with a finger, each hole (24, 10, 2, 5 and 16 mm) sits on the axis without too much play and the wheel does not rub on the wall or the lid.	☐	☐
8	Lid closed with 4 screws, no trapped wire, slack in wires F1 to F3; no oil ran into the box.	☐	☐
9	Nothing loose when shaken gently; no burrs on the printed parts.	☐	☐

Assembler		Date		Inspector	